MILITARY SPECIFICATION

HOSE, SYNTHETIC RUBBER, SYNTHETIC FIBER REINFORCED, AND END FITTINGS, REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES (LOW PRESSURE SIZES -4 THROUGH -32)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers synthetic fiber reinforced hoses and reuseable end fittings in sizes -4 through -32 for use in low pressure applications.

1.2 Classification. End fittings for hose covered by this specification shall be of the following types and nominal sizes:

Type AI - Flange fitting straight, 400 lb/in^2 flange, hose sizes -16 through -32 (see figure 5) (see figure 1 of MIL-H-24136 for flange details).
Type AII - Flange fitting straight, 150/250 lb/in^2 flange, hose size -16 through -32 (see figure 5) (see figure 1 of MIL-H-24136 for flange details).
Type AIII - Flange fitting straight, 250 lb/in^2 flange, hose size -16 through -32 (see figure 5) (see figure 1 of MIL-H-24136 for flange details).
Type AIV - Flange fitting straight, 150 lb/in^2 flange, hose size -16 through -32 (see figure 5) (see figure 1 of MIL-H-24136 for flange details).
Type AV - Flange fitting straight, 150 lb/in^2 B16.5 flange, hose size -16 through -32 (see figure 5) (see figure 1 of MIL-H-24136 for flange details).
Type BI - Flange fitting, 90 degree elbow, 400 lb/in^2 flange, hose size -16 through -32 (see figure 6) (see figure 1 of MIL-H-24136 for flange details).
Type BII - Flange fitting, 90 degree elbow, 150/250 lb/in^2 flange, hose size -16 through -32 (see figure 6) (see figure 1 of MIL-H-24136 for flange details).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4730

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Type BIII - Flange fitting, 90 degree elbow, 250 lb/in^2 flange, hose size -16 through -32 (see figure 6) (see figure 1 of MIL-H-24136 for flange details).
Type BIV - Flange fitting, 90 degree elbow, 150 lb/in^2 flange, hose size -16 through -32 (see figure 6) (see figure 1 of MIL-H-24136 for flange details).
Type BV - Flange fitting, 90 degree elbow, 150 lb/in^2 B16.5 flange, hose size -16 through -32 (see figure 6) (see figure 1 of MIL-H-24136 for flange details).
Type E - 90 degree dogleg elbow, hose sizes -6 to -16 (see figure 4).
Type G - SAE, 37 degree tube swivel to hose, straight, hose sizes -4 to -32 (see figure 1).
Type I - Male half O-ring union, straight, hose sizes -6 through -32 (see figure 2).
Type IL - Male half O-ring union, 90 degree elbow, hose sizes -6 through -32 (see figure 3).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. Unless otherwise specified, the following specifications of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought.
QQ-N-288 - Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.

MILITARY

MIL-H-24136 - Hose, Synthetic Rubber, Synthetic Fiber Reinforced and End Fittings, Reusable, for Flexible Hose Assemblies, General Specification For.
MIL-B-24480 - Bronze, Nickel-Aluminum Castings, for Seawater Service.

(Copies of specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Hose and end fittings furnished under this specification shall conform to the requirements of MIL-H-24136 and as specified herein.

3.2 Dimensions and pressures. Hose dimensions and pressures shall be in accordance with table I.

TABLE I. Dash number, performance characteristics and dimensions.

Dash no. (hose size)	Pressure (lb/in^2)		Hose dimensions (inches)				Hose reinforcement		Minimum bend radius (inches)	Peak impulse pressure (lb/in^2)
	Working (max)	Burst (min)	I.D.		O.D.		Construction	No. of layers		
			Min	Max	Min	Max				
-4	1,000	4,000	0.188	0.214	0.500	0.539	Braided	2	3.00	1500
-5	800	3,200	0.250	0.281	0.562	0.591	Braided	2	3.38	1200
-6	650	2,600	0.313	0.394	0.656	0.659	Braided	2	4.00	975
-8	625	2,500	0.406	0.437	0.748	0.789	Braided	2	4.62	938
-10	600	2,400	0.500	0.339	0.899	0.943	Braided	2	5.50	900
-12	550	2,200	0.625	0.667	1.055	1.101	Braided	2	6.50	825
-16	500	2,000	0.875	0.917	1.203	1.265	Braided	2	7.38	750
-20	450	1,800	0.125	1.172	1.469	1.661	Braided	2	9.00	675
-24	400	1,600	1.575	1.422	1.719	1.781	Braided	2	10.60	600
-32	350	1,400	1.812	1.859	2.172	2.266	Braided	2	16.25	425

3.3 Fitting types and sizes. Fitting types and sizes shall be as shown on figures 1 through 6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-H-24136 and as specified herein.

4.2 Sampling for qualification testing.

4.2.1 Hose. Qualification testing of hose shall consist of hose samples as specified in table II. Samples shall be not less than 15 inches in length and shall be fitted with qualified end fittings from at least two previously qualified suppliers (see 4.2.4).

TABLE II. Number of hose samples for qualification testing.

Hose size (dash no.)	Number of hose samples to be provided
-8	4
* -32	4

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4.2.2 End fittings. Qualification testing of end fittings only, shall consist of 16 end fittings of the same type as specified in table III. End fitting samples shall be installed on eight qualified hose samples, in the sizes listed in table II, from at least two previously qualified hoses (see 4.2.4).

TABLE III. Number of end fitting samples for qualification testing.

End fitting size (hose dash no.)	Number of end fitting samples to be provided
-8	8
-32	8

4.2.3 Hose and end fittings combined. In the event that a single contractor of both hose and fittings desires to qualify both hose and fittings, or a hose contractor and a fitting contractor desire to qualify hose and fittings in a combined qualification test, the samples shall be provided as specified in table IV.

4.2.4 Sample requirements when there are less than two previously qualified suppliers. Where there is only one qualified source for qualified samples all the previously qualified samples required shall come from that source. Where there are no previously qualified sources, all samples of hose and fittings shall be submitted by the prospective qualifier.

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TABLE IV. Number of samples for combined qualification testing.

One approved source					
Hose size (dash no.)	No. of hose samples to be provided		End fitting (Hose dash no.)	No. of end fittings to be provided	
	Qualifier 2/	Approved 1/ source		Qualifier 2/	Approved 1/ source
-8	2	2	-8	4	4
-32	2	2	-32	4	4

Two or more approved sources						
Hose size (dash no.)	Number of samples to be provided by			End fitting size (dash no.)	Number of end fittings to be provided by		
	Qualifier 2/	Approved 1/ source #1	Approved 1/ source #2		Qualifier 2/	Approved 1/ source #1	Approved 1/ source #2
-8	2	1	1	-8	4	2	2
-32	2	1	1	-32	4	2	2

Hose samples shall be not less than 15 inches in length and shall be coupled on one end with the fitting being submitted for qualification and on the other end with previously qualified end fittings.

1/ Approved source - Previously qualified hose and fittings, or hose or fitting.
2/ Qualifier - Hose and fittings, or hose or fitting to be qualified.

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4.2.5 Sequence of tests on sample hose assemblies shall be as specified in table V.

TABLE V. Sequence of qualification tests on sample assemblies.

Test sequence	Sample numbers							
	Size (dash no.) -8				Size (dash no.) -32			
	1	2	3	4	5	6	7	8
Proof pressure (see 4.6.1)1/	X	X	X	X	X	X	X	X
Stability test (see 4.6.2)1/	X	X	X	X	X	X	X	X
Aging test (see 4.6.3)1/		X	X	X		X	X	X
Hydraulic fluid circulation (see 4.6.4)1/		X				X		
Impulse test (see 4.6.5)1/				X				X
Burst test (see 4.6.6)1/	X				X			
Cold temperature flexing (see 4.6.8)1/			X				X	

1/ See MIL-H-24136.

4.2.6 Satisfactory completion of qualification testing of the samples selected in accordance with tables II, III and IV will qualify the manufacturer for all sizes shown in table I.

5. PACKAGING

5.1 Preparation for delivery requirements shall be in accordance with MIL-H-24136.

6. NOTES

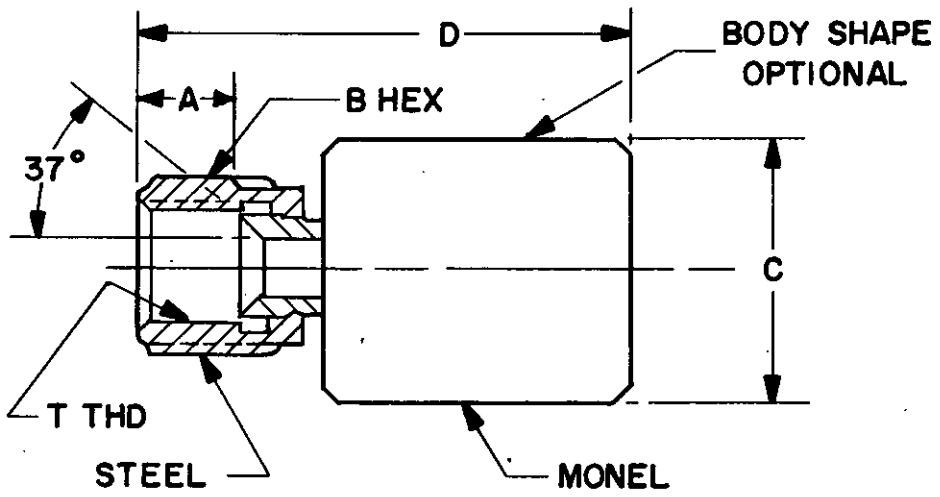
6.1 Intended use. Hose and fittings to this specification sheet are intended for use in shipboard or shore to ship piping systems where pressures do not exceed the rated working pressure and temperatures do not exceed minus 40°F or 200°F.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of end fittings for hoses (see 1.2).

Preparing activity:
Navy - SH
(Project 4730-N874-3)

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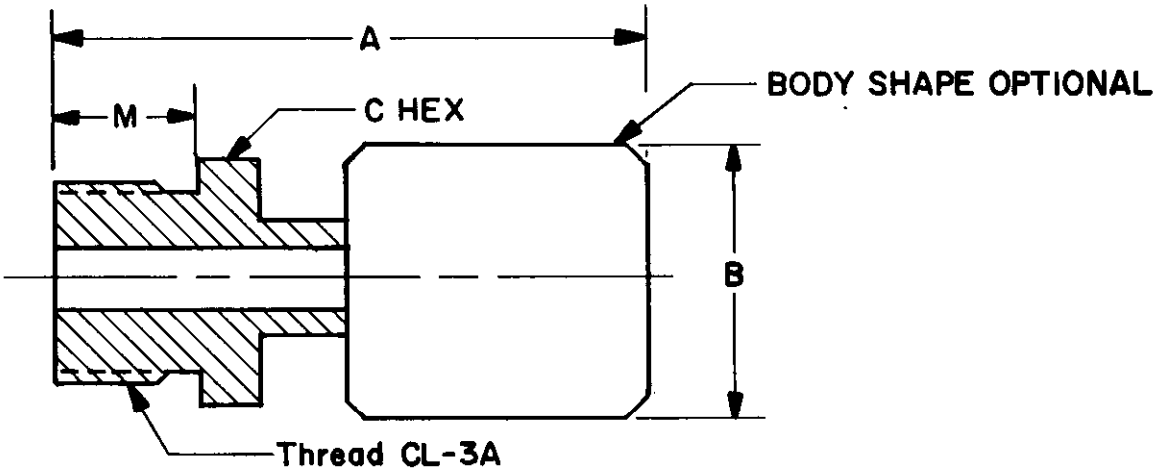
Tube size (nom)	Rubber hose (dash no.)	T thread CL-3	A (max)	B (min)	C (max)	D (max)
1/4	-4	7/16-20	0.35	0.56	0.62	1.75
5/16	-5	1/2-20	0.38	0.62	0.69	1.40
3/8	-6	9/16-18	0.38	0.69	0.81	2.05
1/2	-8	3/4-16	0.43	0.88	0.94	2.52
5/8	-10	7/8-14	0.51	1.00	1.12	2.80
3/4	-12	1-1/16-12	0.57	1.25	1.25	3.15
1	-16	1-5/16-12	0.60	1.50	1.44	2.82
1-1/4	-20	1-5/8-12	0.64	2.00	1.75	3.00
1-1/2	-24	1-7/8-12	0.74	2.25	1.94	2.28
2	-32	2-1/2-12	0.76	2.88	2.44	3.88

SH 12301

NOTES:
1. Dimensions are in inches.
2. Nickel-copper in accordance with QQ-N-281 or nickel-copper-aluminum in accordance with QQ-N-286 is an acceptable substitute for steel.

FIGURE 1. Type G, SAE 37 degree swivel to hose, straight.

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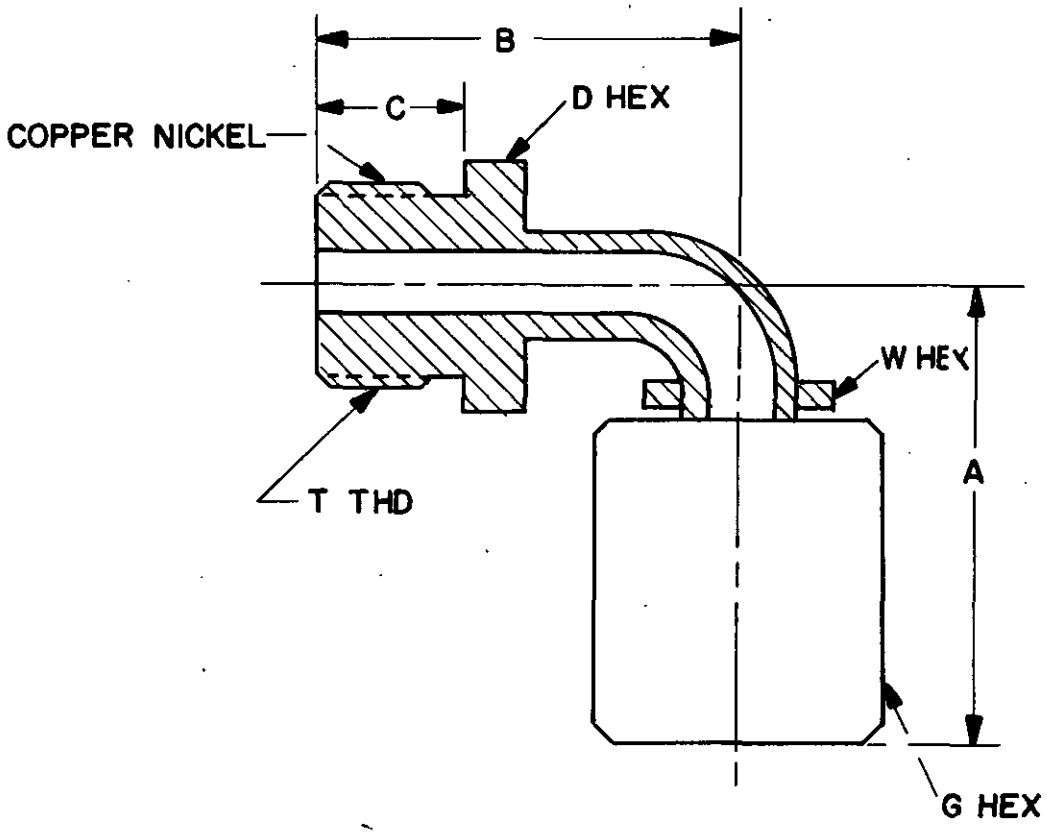
NPS	Thread	Rubber hose (dash no.)	A (max)	B (max)	C (min)	M (min)
1/8	1-14	-6	2.06	0.81	1.06	0.50
1/4	1-3/16-12	-8	2.54	0.94	1.25	0.62
3/8	1-3/8-12	-10	2.75	1.12	1.44	0.62
1/2	1-3/4-12	-12	3.12	1.25	1.81	0.62
3/4	2-12	-16	2.80	1.44	2.12	0.62
1	2-5/16-12	-20	3.14	1.75	2.38	0.75
1-1/4	2-3/4-12	-24	3.23	1.94	2.88	0.75
1-1/2	3-1/16-12	-32	3.89	2.44	3.12	0.88

SH 12302

NOTES:
1. Dimensions are in inches.
2. All parts in direct contact with sea water shall be in accordance with QQ-N-286.
   Bolts and nuts - QQ-N-286.
   All other parts - QQ-N-281, QQ-N-288 or QQ-N-286.

FIGURE 2. Type I, O-ring union straight.

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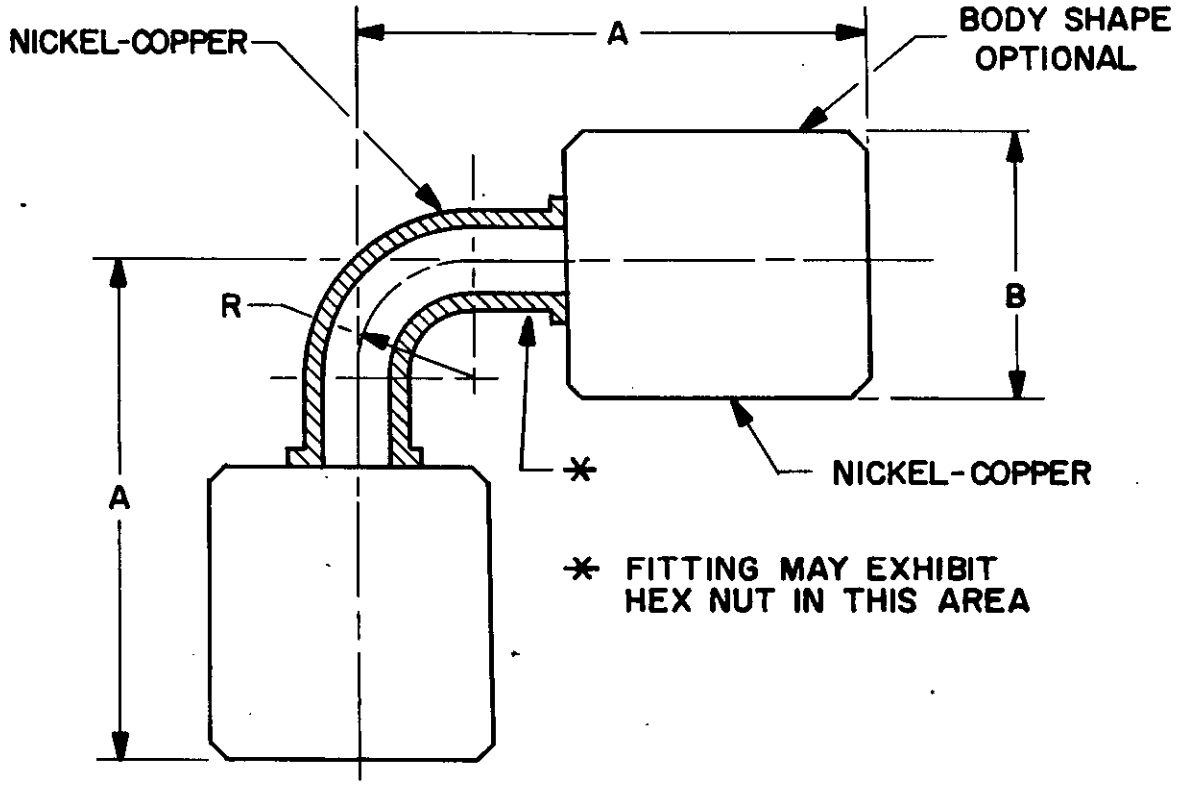
NPS	Rubber hose (dash no.)	T thread	A (max)	B (min)	C (min)	D (min)	G (max)	W (min)
1/8	-6	1-14	2.45	1.30	0.50	1.06	0.81	0.56
1/4	-8	1-3/16-12	3.05	1.68	0.62	1.25	0.94	0.75
3/8	-10	1-3/8-12	3.57	1.99	0.62	1.44	1.12	0.81
1/2	-12	1-3/4-12	4.15	2.31	0.62	1.81	1.25	1.00
3/4	-16	2-12	3.93	2.68	0.62	2.12	1.44	1.38
1	-20	2-5/16-12	4.27	3.17	0.75	2.38	1.75	1.62
1-1/4	-24	2-3/4-12	4.65	3.42	0.75	2.88	1.94	1.81
1-1/2	-32	3-1/16-12	5.68	4.05	0.88	3.12	2.44	2.38

SH 12303

NOTES:
1. Dimensions are in inches.

FIGURE 3. Type IL, 90 degree O-ring union.

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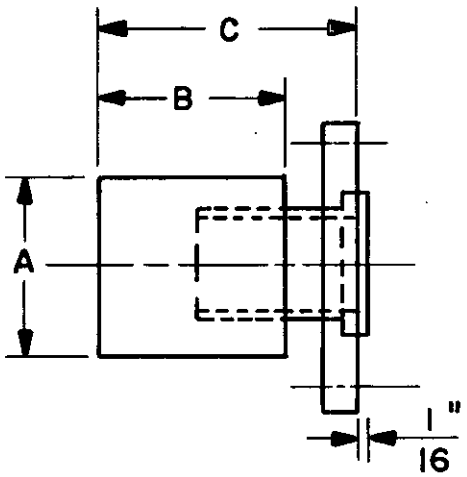
Hose size (dash no.)	A (max)	B (max)	R (nom)
-6	2.45	0.81	0.50
-8	3.05	0.94	0.75
-10	3.57	1.12	1.00
-12	4.15	1.25	1.25
-16	3.93	1.44	1.50
-20	2.98	1.75	1.50
-24	3.24	1.94	2.00
-32	3.28	2.44	2.50

SH 12306

NOTES:
1. Dimensions are in inches.

FIGURE 4. Type E, 90 degree elbow, hose to hose.

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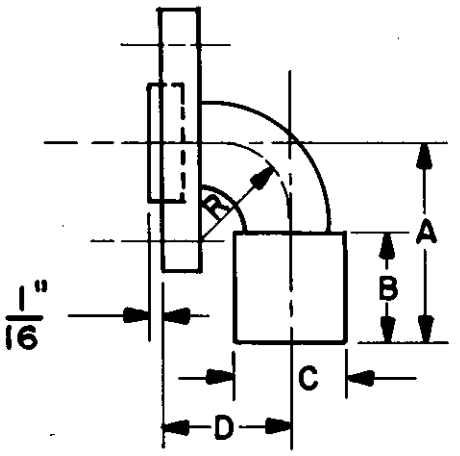
NPS	Rubber hose (dash no.)	A (max)	B (max)	C (max)
1/2	-16	1.44	1.66	2.80
3/4	-20	1.75	1.75	2.89
1	-24	1.94	1.84	2.92
1-1/4	-32	2.44	2.25	3.51
1-1/2	-32	2.44	2.25	3.51
2	-32	2.44	2.25	3.51

SH 12305

NOTES:
1. Dimensions are in inches.
2. All parts in direct contact with sea water shall be in accordance with MIL-B-24480.
   Bolts and nuts - QQ-N-286.
   All other parts - QQ-N-281, QQ-N-288, MIL-B-24480 or QQ-N-286.

FIGURE 5. Types AI, AII, AIII, AIV or AV, end fitting, hose to flange, straight.

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NPS	Hose size	A (max)	B (max)	C (max)	D (nom)	R (nom)
1/2	-16	2.80	1.66	1.53	2.88	1.12
3/4	-20	2.98	1.75	1.62	2.56	1.82
1	-24	3.24	1.84	1.78	2.75	1.28
1-1/4	-32	3.89	2.25	2.05	3.06	1.62
1-1/2	-32	3.89	2.25	2.05	3.06	1.62
2	-32	3.89	2.25	2.05	3.06	1.62

SH 12306

NOTES:
1. Dimensions are in inches.
2. All parts in direct contact with sea water shall be in accordance with MIL-B-24480.
   Bolts and nuts - QQ-N-286.
   All other parts - QQ-N-281, QQ-N-288, MIL-B-24480 or QQ-N-286.

FIGURE 6. Types BI, BII, BIII, BIV, or BV, end fitting, flange to hose, elbow sizes -16 through -32.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-24136/3(SH)	Hose, Synthetic Rubber, Synthetic Fiber Reinforced And End Fittings, Reusable, For Flexible Hose Assemblies

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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